

**DATABASE MANAGEMENT SYSTEM : DBMS
PROJECT**

DELIVERY MANAGEMENT SYSTEM

Main Report

By

OBLI SANJAY S 717823D140

SANDEEP KUMAR P H 717823D147

Github link of the project- https://github.com/sandeepkumarph/DBMS_PROJECT.git



DEC - 2025

TABLE OF CONTENTS:

S. NO	CONTENT
1	ER DIAGRAM
2	SQL CODE WITH SCREENSHOTS
3	QUERY RESULTS WITH EXPLANATIONS
4	EXECUTION PLANS (INDEX VS NO INDEX)
5	TRANSACTION & RECOVERY DEMONSTRATIONS
6	SECURITY ROLE ENFORCEMENT & SQL INJECTION DEMO

ER DIAGRAM:

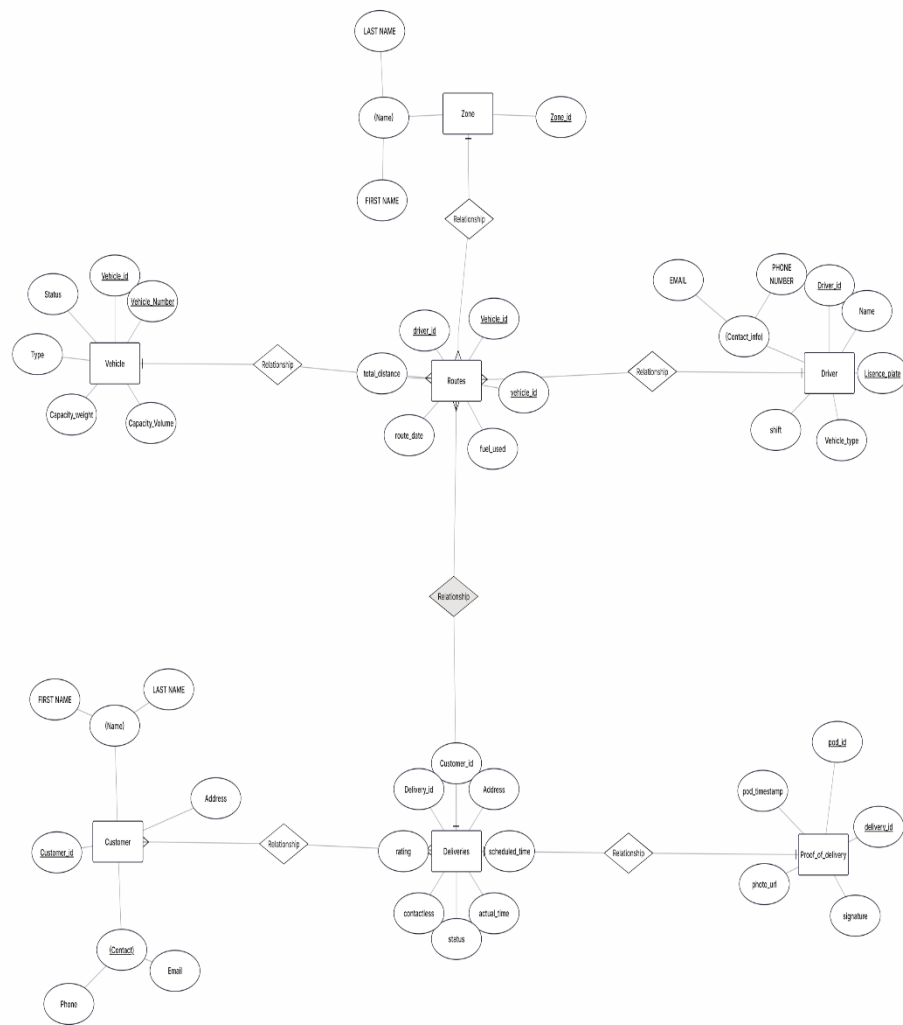


TABLE CREATION:

1.driver

2.vehicle

3.zone

The screenshot shows the MySQL Workbench interface with a query window containing the following SQL code:

```
1 create database DeliveryManagementSystem;
2 use DeliveryManagementSystem;
3 CREATE TABLE driver (
4     driver_id INT PRIMARY KEY,
5     first_name VARCHAR(50) NOT NULL,
6     license_number VARCHAR(20) NOT NULL UNIQUE,
7     phone VARCHAR(20),
8     shift VARCHAR(20)
9 );
10 create TABLE vehicle (
11     vehicle_id INT PRIMARY KEY,
12     license_plate VARCHAR(20) NOT NULL UNIQUE,
13     vehicle_type VARCHAR(20) NOT NULL,
14     weight_capacity INT,
15     volume_capacity INT
16 );
17 CREATE TABLE zone (
18     zone_id INT PRIMARY KEY,
19     zone_name VARCHAR(50) ,
20     description VARCHAR(255) not null unique
21 );
```

The output window shows the execution results:

#	Time	Action	Message	Duration / Fetch
1	19:25:58	create database DeliveryManagementSystem	1 row(s) affected	0.015 sec
2	19:26:58	use DeliveryManagementSystem	0 row(s) affected	0.000 sec
3	19:28:41	CREATE TABLE driver (driver_id INT PRIMARY KEY, first_name VARCHAR(50) NOT NULL, license_number ...	0 row(s) affected	0.062 sec
4	19:31:55	create TABLE vehicle (vehicle_id INT PRIMARY KEY, license_plate VARCHAR(20) NOT NULL UNIQUE, ...	Error Code: 1064. You have an error in your SQL syntax; check the manual that corresponds to your MySQL server v...	0.000 sec
5	19:32:10	create TABLE vehicle (vehicle_id INT PRIMARY KEY, license_plate VARCHAR(20) NOT NULL UNIQUE, ve...	0 row(s) affected	0.076 sec
6	19:33:28	CREATE TABLE zone (zone_id INT PRIMARY KEY, zone_name VARCHAR(50) NOT NULL UNIQUE, desc...	0 row(s) affected	0.063 sec

4.routes

5.customer

The screenshot shows the MySQL Workbench interface with a query window containing the following SQL code:

```
22 CREATE TABLE routes (
23     route_id INT PRIMARY KEY,
24     route_date DATE NOT NULL,
25     driver_id INT NOT NULL,
26     vehicle_id INT,
27     total_distance INT,
28     CONSTRAINT fk_routes_driver FOREIGN KEY (driver_id) REFERENCES Driver(driver_id),
29     CONSTRAINT fk_routes_vehicle FOREIGN KEY (vehicle_id) REFERENCES Vehicle(vehicle_id)
30 );
31 CREATE TABLE customer (
32     customer_id INT PRIMARY KEY,
33     name VARCHAR(100) NOT NULL,
34     address VARCHAR(255),
35     city VARCHAR(50),
36     state VARCHAR(50),
37     zip_code VARCHAR(10),
38     phone VARCHAR(20),
39     email VARCHAR(100) UNIQUE,
40     zone_id INT,
41     CONSTRAINT fk_customer_zone FOREIGN KEY (zone_id) REFERENCES Zone(zone_id)
42 );
```

The output window shows the execution results:

#	Time	Action	Message	Duration / Fetch
6	19:33:28	CREATE TABLE zone (zone_id INT PRIMARY KEY, zone_name VARCHAR(50) NOT NULL UNIQUE, de...	0 row(s) affected	0.063 sec
7	19:37:51	CREATE TABLE routes (route_id INT PRIMARY KEY, route_date DATE NOT NULL, driver_id INT NOT NU...	0 row(s) affected	0.063 sec
8	19:39:19	CREATE TABLE customer (customer_id INT PRIMARY KEY, name VARCHAR(100) NOT NULL, address V...	Error Code: 1064. You have an error in your SQL syntax; check the manual that corresponds to your MySQL server ...	0.000 sec
9	19:39:47	CREATE TABLE customer (customer_id INT PRIMARY KEY, name VARCHAR(100) NOT NULL, address ...	Error Code: 1064. You have an error in your SQL syntax; check the manual that corresponds to your MySQL server ...	0.000 sec
10	19:40:26	CREATE TABLE routes (route_id INT PRIMARY KEY, route_date DATE NOT NULL, driver_id INT NOT NU...	Error Code: 1050. Table 'routes' already exists	0.000 sec
11	19:40:50	CREATE TABLE customer (customer_id INT PRIMARY KEY, name VARCHAR(100) NOT NULL, address ...	0 row(s) affected	0.047 sec

7.proof or delivery

The screenshot shows the MySQL Workbench interface with the 'delivery_management_system' database selected. The 'Query 1' window contains the following SQL code:

```
43 CREATE TABLE deliveries (
44     delivery_id INT PRIMARY KEY,
45     route_id INT NOT NULL,
46     customer_id INT NOT NULL,
47     zone_id INT NOT NULL,
48     delivery_type VARCHAR(20),
49     status VARCHAR(20),
50     scheduled_time DATETIME,
51     actual_time DATETIME,
52     attempt_count INT,
53     is_returned BOOLEAN DEFAULT FALSE
54 );
55 CREATE TABLE proof_of_delivery (
56     proof_id INT PRIMARY KEY,
57     delivery_id INT UNIQUE,
58     signature BLOB,
59     photo BLOB,
60     delivered_at DATETIME NOT NULL
61 );
```

The 'Output' window shows the execution results:

#	Time	Action	Message	Duration / Fetch
12	19:48:25	CREATE TABLE deliveries (delivery_id INT PRIMARY KEY, route_id INT NOT NULL, customer_id INT N...	Error Code: 1064. You have an error in your SQL syntax; check the manual that corresponds to your MySQL server ...	0.016 sec
13	19:48:40	CREATE TABLE deliveries (delivery_id INT PRIMARY KEY, route_id INT NOT NULL, customer_id INT N...	0 row(s) affected	0.031 sec
14	19:51:56	CREATE TABLE proof_of_delivery (proof_id INT PRIMARY KEY, delivery_id INT UNIQUE, signature BLO...	0 row(s) affected	0.047 sec

INSERTING VALUES TO THE TABLES:

- 1.driver
- 2.vehicles
- 3.zone

The screenshot shows the MySQL Workbench interface with the 'delivery_management_system' database selected. The 'Query 1' window contains the following SQL code:

```
63 #Inserting values into driver table
64 Insert into driver values
65 (101, 'Ramesh', 'DL-12345', '1234567890', 'Morning'),
66 (102, 'Suresh', 'DL-54321', '5432167890', 'Evening'),
67 (103, 'Rajesh', 'DL-67890', '0987654321', 'Afternoon'),
68 (104, 'Sanjay', 'DL-11111', '0987654321', 'Evening');
69
70 #Inserting values into vehicle table
71 INSERT INTO vehicle values
72 (201, 'TN36AB1234', 'Bike', 150, 1),
73 (202, 'TN36CD5678', 'Van', 800, 6),
74 (203, 'TN36EF9012', 'Truck', 1000, 18),
75 (204, 'TN99GH3456', 'Van', 800, 5),
76 (205, 'TN99IJ7890', 'Bike', 200, 1);
77
78 #Inserting values into zone table
79 INSERT into zone values
80 (1, 'Zone 1', 'Residential area'),
81 (2, 'Zone 2', 'Village area'),
82 (3, 'Zone3', 'City area'),
83 (4, 'Zone4', 'Rural area'),
84 (5, 'Zone5', 'Urban area');
85
```

The 'Output' window shows the execution results:

#	Time	Action	Message	Duration / Fetch
15	20:00:46	Insert into driver values (101, 'Ramesh', 'DL-12345', '1234567890', 'Morning'), (102, 'Suresh', 'DL-54321', '54321678...	4 row(s) affected Records: 4 Duplicates: 0 Warnings: 0	0.000 sec
16	20:03:12	INSERT INTO vehicle values (201, 'TN36AB1234', 'Bike', 150, 1), (202, 'TN36CD5678', 'Van', 800, 6), (203, 'TN3...	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0	0.016 sec
17	20:07:51	INSERT into zone values (1, 'Zone 1', 'Residential area'), (2, 'Zone 2', 'Village area'), (3, 'Zone3', 'City area'), (4, 'Zon...	Error Code: 1064. You have an error in your SQL syntax; check the manual that corresponds to your MySQL server ...	0.000 sec
18	20:08:04	INSERT into zone values (1, 'Zone 1', 'Residential area'), (2, 'Zone 2', 'Village area'), (3, 'Zone3', 'City area'), (4, 'Zs...	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0	0.016 sec

4.routes

5.customer

6.delivery

MySQL Workbench

delivery_management_system x

File Edit View Query Database Server Tools Scripting Help

Navigator

Filter objects

SCHEMAS

deliverymanagementsystem

Tables

- deliveries
- customer
- driver
- proof_of_delivery
- routes
- vehicle
- zone

Views

Stored Procedures

Functions

sakila

sys

world

Administration Schemas

Information

Schema: deliverymanagementsystem

Query 1

```
#Inserting values into routes table
85
86 • INSERT INTO routes VALUES
87 (301, '2025-03-10', 101, 201, 45),
88 (302, '2025-03-10', 102, 202, 60),
89 (303, '2025-03-11', 103, 203, 120),
90 (304, '2025-03-11', 104, 204, 55),
91 (305, '2025-03-12', 101, 205, 40);
92
93 #Inserting values into customer table
94
95 (1, 'Ramesh raj', 'md colony', 'Chennai', 'TN', '600001', '9000011111', 'ramesh@gmail.com',
96 (2, 'Suresh kumar', 'CS nagan', 'Chennai', 'TN', '600002', '9000022222', 'suresh@gmail.com',
97 (3, 'Anitha', 'Eachanari', 'Coimbatore', 'TN', '641001', '9000033333', 'anitha@gmail.com',
98 (4, 'Kumar', 'Vercaud', 'Salem', 'TN', '636001', '9000044444', 'kumar@gmail.com', 4),
99 (5, 'Divya', 'Anna Nagar', 'Madurai', 'TN', '625001', '9000055555', 'divya@gmail.com', 5);
100
101 #Inserting values into deliveries table
102
103 (1, 301, 1, 1, 'Drop', 'Delivered', '2025-03-10', '2025-03-10', 1, FALSE),
104 (2, 302, 2, 2, 'Drop', 'Failed', '2025-03-10', '2025-03-10', 2, TRUE),
105 (3, 303, 3, 3, 'Pickup', 'Delivered', '2025-03-11', '2025-03-11', 1, FALSE),
106 (4, 304, 4, 4, 'Drop', 'Failed', '2025-03-11', '2025-03-11', 2, FALSE),
107 (5, 305, 5, 5, 'Drop', 'Delivered', '2025-03-12', '2025-03-12', 1, FALSE);
108
109
110
111
112
113
114
115
116
117
118
119
120
121
122
123
124
125
126
127
128
129
130
131
132
```

SQLAdditions

Jump to

Output

Action Output

#	Time	Action	Message	Duration / Fetch
22	20:19:46	INSERT INTO routes VALUES (301, '2025-03-10', 101, 201, 45), (302, '2025-03-10', 102, 202, 60), (303, '2025-03-11', 103, 203, 120), (304, '2025-03-11', 104, 204, 55), (305, '2025-03-12', 101, 205, 40);	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0	0.000 sec
23	20:27:14	INSERT INTO customer values (1, 'Ramesh raj', 'md colony', 'Chennai', 'TN', '600001', '9000011111', 'ramesh@gmail.com', 'md colony', 'Chennai', 'TN', '600001', '9000011111', 'ramesh@gmail.com');	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0	0.016 sec
24	20:34:08	INSERT INTO deliveries (1, 301, 1, 1, 'Drop', 'Delivered', '2025-03-10', '2025-03-10', 1, FALSE), (2, 302, 2, 2, 'Drop', 'Failed', '2025-03-10', '2025-03-10', 2, TRUE), (3, 303, 3, 3, 'Pickup', 'Delivered', '2025-03-11', '2025-03-11', 1, FALSE), (4, 304, 4, 4, 'Drop', 'Failed', '2025-03-11', '2025-03-11', 2, FALSE), (5, 305, 5, 5, 'Drop', 'Delivered', '2025-03-12', '2025-03-12', 1, FALSE);	Error Code: 1064. You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near 'md colony', 'Chennai', 'TN', '600001', '9000011111', 'ramesh@gmail.com', 'md colony', 'Chennai', 'TN', '600001', '9000011111', 'ramesh@gmail.com' at line 1	0.000 sec
25	20:34:25	INSERT INTO deliveries VALUES (1, 301, 1, 1, 'Drop', 'Delivered', '2025-03-10', '2025-03-10', 1, FALSE), (2, 302, 2, 2, 'Drop', 'Failed', '2025-03-10', '2025-03-10', 2, TRUE), (3, 303, 3, 3, 'Pickup', 'Delivered', '2025-03-11', '2025-03-11', 1, FALSE), (4, 304, 4, 4, 'Drop', 'Failed', '2025-03-11', '2025-03-11', 2, FALSE), (5, 305, 5, 5, 'Drop', 'Delivered', '2025-03-12', '2025-03-12', 1, FALSE);	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0	0.016 sec

Query Completed

7.proof of delivery

MySQL Workbench

delivery_management_system x

File Edit View Query Database Server Tools Scripting Help

Navigator

Filter objects

SCHEMAS

deliverymanagementsystem

Tables

- deliveries
- customer
- driver
- proof_of_delivery
- routes
- vehicle
- zone

Views

Stored Procedures

Functions

sakila

sys

world

Administration Schemas

Information

Schema: deliverymanagementsystem

Query 1

```
#INSERTING VALUES INTO PROOF OF DEIVERY TABLE
107
108
109
110
111
112
113
114 • INSERT INTO proof_of_delivery VALUES
115 (1, 1, NULL, NULL, '2025-03-10'),
116 (2, 2, NULL, NULL, '2025-03-10'),
117 (3, 3, NULL, NULL, '2025-03-11'),
118 (4, 4, NULL, NULL, '2025-03-11'),
119 (5, 5, NULL, NULL, '2025-03-12');
120
121
122
123
124
125
126
127
128
129
130
131
132
```

SQLAdditions

Jump to

Output

Action Output

#	Time	Action	Message	Duration / Fetch
25	20:34:25	INSERT INTO deliveries VALUES (1, 301, 1, 1, 'Drop', 'Delivered', '2025-03-10', '2025-03-10', 1, FALSE), (2, 302, 2, 2, 'Drop', 'Failed', '2025-03-10', '2025-03-10', 2, TRUE), (3, 303, 3, 3, 'Pickup', 'Delivered', '2025-03-11', '2025-03-11', 1, FALSE), (4, 304, 4, 4, 'Drop', 'Failed', '2025-03-11', '2025-03-11', 2, FALSE), (5, 305, 5, 5, 'Drop', 'Delivered', '2025-03-12', '2025-03-12', 1, FALSE);	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0	0.016 sec

Query Completed

SQL queries:

1. List all drivers who drive "Vans".

The screenshot shows the MySQL Workbench interface with the 'delivery_management_system' database selected. The 'Schemas' pane on the left shows the database structure, including tables like 'customer', 'deliveries', 'driver', 'proof_of_delivery', 'routes', and 'vehicle'. The 'Query' editor in the center contains the following SQL query:

```
120
121 use deliverymanagementsystem;
122
123 select * from vehicle ;
124 select * from vehicle where vehicle_type='Van';
125
126
```

The 'Result Grid' shows the results of the query, displaying columns: vehicle_id, license_plate, vehicle_type, weight_capacity, and volume_capacity. The results are as follows:

vehicle_id	license_plate	vehicle_type	weight_capacity	volume_capacity
202	TN36CD5678	Van	800	6
204	TN99G+3456	Van	800	5

The 'Output' pane at the bottom shows the execution details of the query, including the time taken and the number of rows returned.

2. Find deliveries scheduled for "Zone 1".

The screenshot shows the MySQL Workbench interface with the 'delivery_management_system' database selected. The 'Schemas' pane on the left shows the database structure, including tables like 'customer', 'deliveries', 'driver', 'proof_of_delivery', 'routes', and 'vehicle'. The 'Query' editor in the center contains the following SQL query:

```
126
127
128
129
130 select * from deliveries;
131 select * FROM deliveries where zone_id='1';
132
```

The 'Result Grid' shows the results of the query, displaying columns: delivery_id, route_id, customer_id, zone_id, delivery_type, status, scheduled_time, and actual_time. The results are as follows:

delivery_id	route_id	customer_id	zone_id	delivery_type	status	scheduled_time	actual_time
1	301	1	1	Drop	Delivered	2025-03-10 00:00:00	2025-03-10 00:00:00

The 'Output' pane at the bottom shows the execution details of the query, including the time taken and the number of rows returned.

3. Retrieve deliveries marked as "Failed Attempt".

The screenshot shows the MySQL Workbench interface with the 'delivery_management_system' database selected. The 'Schemas' pane on the left shows the database structure, including tables like 'customer', 'deliveries', 'driver', 'proof_of_delivery', 'routes', 'vehicle', and 'zone'. The 'Query' editor in the center contains the following SQL query:

```
132
133 select * from deliveries;
134 select * from deliveries WHERE status = 'failed';
135
136
137
138
```

The 'Result Grid' shows the results of the query, displaying columns: delivery_id, route_id, customer_id, zone_id, delivery_type, status, scheduled_time, actual_time, and attempt. The results show two failed deliveries:

delivery_id	route_id	customer_id	zone_id	delivery_type	status	scheduled_time	actual_time	attempt
2	302	2	2	Drop	Failed	2025-03-10 00:00:00		2
4	304	4	4	Drop	failed	2025-03-11 00:00:00		0

The 'Output' pane at the bottom shows the execution progress:

#	Time	Action	Message	Duration / Fetch
1	09:51:04	select * from deliveries LIMIT 0, 1000	5 row(s) returned	0.000 sec / 0.000 sec
2	09:53:24	select * from deliveries WHERE status = 'failed' LIMIT 0, 1000	2 row(s) returned	0.000 sec / 0.000 sec

4. List vehicles with a capacity greater than 500kg.

The screenshot shows the MySQL Workbench interface with the 'delivery_management_system' database selected. The 'Schemas' pane on the left shows the database structure, including tables like 'customer', 'deliveries', 'driver', 'proof_of_delivery', 'routes', 'vehicle', and 'zone'. The 'Query' editor in the center contains the following SQL query:

```
138
139
140
141
142 select * from vehicle;
143 select * from vehicle where weight_capacity>500;
144
```

The 'Result Grid' shows the results of the query, displaying columns: vehicle_id, license_plate, vehicle_type, weight_capacity, and volume_capacity. The results show three vehicles:

vehicle_id	license_plate	vehicle_type	weight_capacity	volume_capacity
202	TH36CD5678	Van	800	6
203	TH38PH012	Truck	1000	18
204	TH99GH456	Van	800	5

The 'Output' pane at the bottom shows the execution progress:

#	Time	Action	Message	Duration / Fetch
1	09:54:18	select * from vehicle LIMIT 0, 1000	5 row(s) returned	0.000 sec / 0.000 sec
2	09:55:18	select * from vehicle where capacity>500 LIMIT 0, 1000	Error Code: 1054. Unknown column 'capacity' in 'where clause'	0.015 sec
3	09:55:56	select * from vehicle where weight_capacity>500 LIMIT 0, 1000	3 row(s) returned	0.000 sec / 0.000 sec

5. Find customers who have requested "Contactless Delivery".

The screenshot shows the MySQL Workbench interface. The left sidebar displays the database schema for 'delivery_management_system'. The main query editor contains the following SQL code:

```
157 select * from deliveries;
158 insert into deliveries values
159 (6, 303, 3, 3, 'contactless_delivery', 'delivered', '2025-03-13', NULL, 1, FALSE);
160 select * from customer;
161 select * from customer c join deliveries d
162 ON c.customer_id=d.customer_id
163 where d.delivery_type='contactless_delivery';
```

The 'Result Grid' shows the output of the query, displaying columns: customer_id, name, address, city, state, zip_code, phone, email, zone_id, and delivery_type. The first row shows a customer named 'Anitha' with a 'contactless_delivery' status.

The 'Output' tab at the bottom shows the execution log, including the message: 'Error Code: 1136. Column count doesn't match value count at row 1'.

Joins & Subqueries

6. Display delivery details along with Driver Name and Vehicle Number.

The screenshot shows the MySQL Workbench interface. The left sidebar displays the database schema for 'delivery_management_system'. The main query editor contains the following SQL code:

```
181 select * from deliveries ;
182 select * from routes;
183 select * from driver;
184 alter table deliveries add driver_id int;
185 update deliveries
186 set driver_id=102
187 where delivery_id in (2,11);
188 select dv.first_name as driver_name,v.license_plate AS vehicle_number,d.delivery_id,d.customer_id
189 from deliveries d
190 join driver dv
191 on d.driver_id=dv.driver_id
192 join routes r on
193 d.route_id=r.route_id
194 join vehicle v
195 on r.vehicle_id=v.vehicle_id
```

The 'Result Grid' shows the output of the query, displaying columns: driver_name, vehicle_number, delivery_id, customer_id, and status. The first two rows show deliveries for 'Ramesh' and 'Suresh'.

The 'Output' tab at the bottom shows the execution log, including the message: 'Error Code: 1136. Column count doesn't match value count at row 1'.

7. Find drivers who have completed deliveries in "Downtown" today.

The screenshot shows the MySQL Workbench interface with a query window titled 'deliveryMS source sdi'. The query is as follows:

```
188
189 select * from deliveries;
190 select * from zone;
191 select * from driver;
192 UPDATE zone
193 SET zone_name = 'Downtown'
194 WHERE zone_id = 3;
195 insert into zone values (6,'down-town','city');
196 insert into zone values (7,'lighttown','city');
197 insert into zone values (8,'racecourse','city');
198 Select dv.driver_id, dv.first_name FROM deliveries d
199 JOIN driver dv
200 on d.driver_id = dv.driver_id
201 JOIN zone z
202 on d.zone_id = z.zone_id
203 WHERE z.zone_name = 'down-town' ;
```

The result grid shows the following data:

driver_id	first_name
103	Rajesh

The output window shows the execution log with the following messages:

#	Time	Action	Message	Duration / Fetch
13	12:07:11	UPDATE zone SET zone_name = 'Downtown' WHERE zone_id = 3	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0	0.000 sec
14	12:07:18	Select dv.driver_id, dv.first_name FROM deliveries d JOIN driver dv on d.driver_id = dv.driver_id JOIN z...	0 row(s) returned	0.000 sec / 0.000 sec
15	12:07:50	select * from driver LIMIT 0, 1000	4 row(s) returned	0.016 sec / 0.000 sec
16	12:07:56	select * from zone LIMIT 0, 1000	8 row(s) returned	0.000 sec / 0.000 sec
17	12:08:16	select * from deliveries LIMIT 0, 1000	10 row(s) returned	0.016 sec / 0.000 sec
18	12:09:00	Select dv.driver_id, dv.first_name FROM deliveries d JOIN driver dv on d.driver_id = dv.driver_id JOIN z...	0 row(s) returned	0.000 sec / 0.000 sec

8. List routes that have more than 50 stops.

The screenshot shows the MySQL Workbench interface with a query window titled 'delivery M5 source sdi'. The query is as follows:

```
207
208
209
210
211
212
217 select * from deliveries;
218 SELECT route_id, COUNT(delivery_id)
219 FROM deliveries
220 GROUP BY route_id
221 HAVING COUNT(delivery_id) > 50;
```

The result grid shows the following data:

route_id	COUNT(delivery_id)
305	55

The output window shows the execution log with the following messages:

#	Action	Message	Duration / Fetch
11:44:01	SELECT * from deliveries LIMIT 0, 1000	1 row(s) returned	0.032 sec / 0.000:
11:44:07	SELECT route_id, COUNT(delivery_id) FROM deliveries GROUP BY route_id HAVING COUNT(delivery_id) > 50;	0.031 sec / 0.000:	
11:44:07	SELECT route_id, COUNT(delivery_id) FROM deliveries GROUP BY route_id HAVING COUNT(delivery_id) > 50;	0.000 sec / 0.000:	

9. Identify deliveries that were reassigned to a different driver.

The screenshot shows the MySQL Workbench interface with a query editor and a result grid. The query is as follows:

```
234 select * from deliveries ;
235 select * from routes;
236 UPDATE deliveries
237 SET driver_id = 181
238 WHERE delivery_id IN (5, 7, 21, 22, 23);
239 UPDATE deliveries
240 SET driver_id = 182
241 WHERE delivery_id = 21;
242 SELECT
243     d.delivery_id,
244     d.route_id,
245     r.driver_id AS original_driver,
246     d.driver_id AS reassigned_driver
247 FROM deliveries d
248 JOIN routes r
249 ON d.route_id = r.route_id
250 WHERE d.driver_id <> r.driver_id;
```

The result grid shows the following data:

delivery_id	route_id	original_driver	reassigned_driver
21	305	101	102

The output pane shows the following action output:

#	Time	Action	Message	Duration / Fetch
19	12:32:16	SELECT d.delivery_id, d.route_id, r.driver_id AS original_driver, d.driver_id AS reassigned_driver	0 row(s) returned	0.000 sec / 0.000 sec
20	12:32:35	select * from deliveries LIMIT 0, 1000	10 row(s) returned	0.016 sec / 0.000 sec
21	12:32:41	select * from routes LIMIT 0, 1000	5 row(s) returned	0.015 sec / 0.000 sec

10. List customers who have provided a "Low Rating" for delivery experience.

The screenshot shows the MySQL Workbench interface with a query editor and a result grid. The query is as follows:

```
283
284
285
286
287
288
289
290 SELECT * FROM customer;
291 SELECT * FROM deliveries;
292 SELECT customer_id, name FROM customer
293 WHERE customer_id in (
294     SELECT customer_id
295     FROM deliveries
296     WHERE rating = 'Low'
297 );
298
299
300
```

The result grid shows the following data:

customer_id	name
1	Ramesh raj

The output pane shows the following action output:

#	Time	Action	Message	Duration / Fetch
38	14:01:45	SELECT * FROM deliveries LIMIT 0, 1000	10 row(s) returned	0.000 sec / 0.000 sec
39	14:01:57	SELECT customer_id, name FROM customer WHERE customer_id in (SELECT customer_id FROM deliveries WHERE rating = 'Low')	1 row(s) returned	0.016 sec / 0.000 sec

Aggregation & Reports

1. Top 5 drivers by delivery completion rate.

The screenshot shows the MySQL Workbench interface with a query window titled 'deliveryMS source sq'. The query is as follows:

```
290 select * from driver;
291
292 SELECT
293   dr.driver_id,
294   dr.first_name AS driver_name, COUNT(CASE WHEN d.status = 'Delivered' THEN 1 END) AS completed
295 FROM deliveries d
296 JOIN driver dr on d.driver_id = dr.driver_id
297 GROUP BY dr.driver_id, dr.first_name
298 ORDER BY completed desc;
```

The left sidebar shows the 'Schemas' pane with the 'deliverymanagementsystem' database selected. The 'Table: deliveries' is highlighted, showing its columns: delivery_id (int PK), customer_id (int), zone_id (int), delivery_type (varchar(20)), status (varchar(20)), scheduled_time (datetime), actual_time (datetime), attempt_count (int), and is_returned (tinyint(1)).

The 'Result Grid' shows the following data:

driver_id	driver_name	completed
101	Ramesh	4
103	Rajesh	2
102	Suresh	1

The 'Action Output' pane shows the execution of the query, indicating that 3 rows were returned.

2. Total distance covered by the fleet today.

The screenshot shows the MySQL Workbench interface with a query window titled 'deliveryMS source sq'. The query is as follows:

```
302
303
304
305
306
307
308
309 update routes set route_date = '2025-12-16';
310 SELECT SUM(total_distance)
311 FROM routes
312 WHERE route_date = current_date;
```

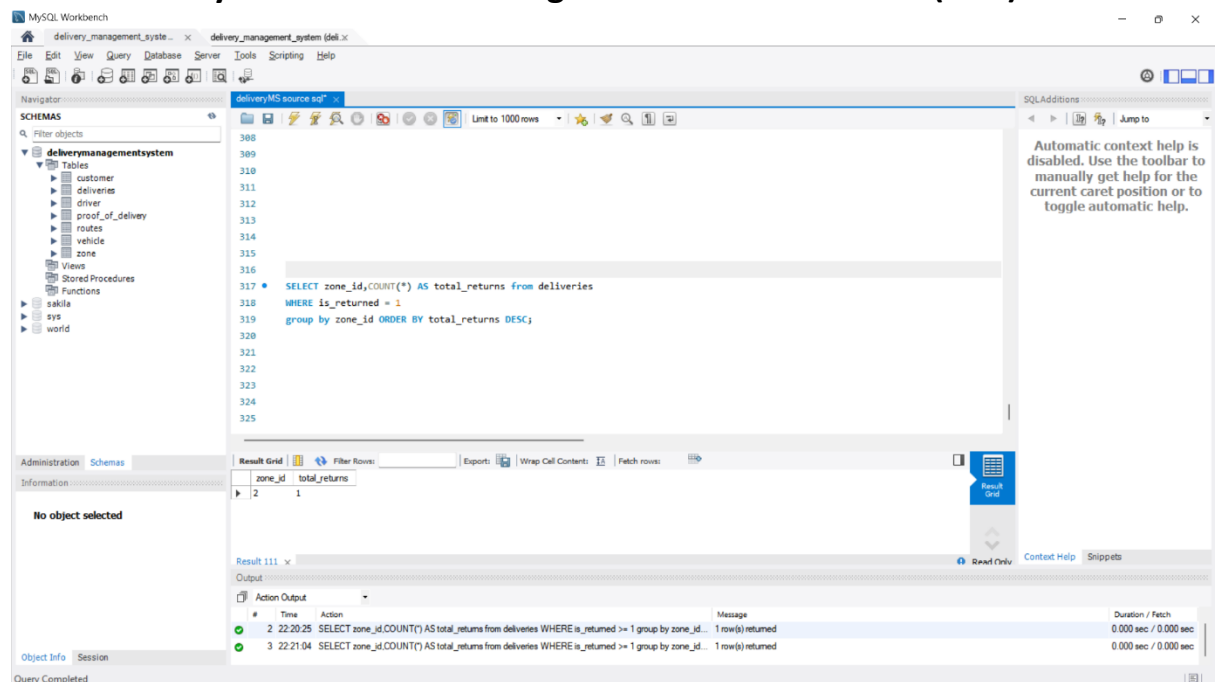
The left sidebar shows the 'Schemas' pane with the 'deliverymanagementsystem' database selected. The 'Table: routes' is highlighted, showing its columns: route_id (int PK), driver_id (int), customer_id (int), zone_id (int), delivery_type (varchar(20)), status (varchar(20)), scheduled_time (datetime), actual_time (datetime), attempt_count (int), and is_returned (tinyint(1)).

The 'Result Grid' shows the following data:

SUM(total_distance)
320

The 'Action Output' pane shows the execution of the query, indicating that 1 row was returned.

3. Identify the zone with the highest number of returns (RTO).



The screenshot shows the MySQL Workbench interface. The left sidebar displays the 'Schemas' tree with 'deliverymanagementsystem' expanded, showing tables like 'customer', 'deliveries', 'driver', 'proof_of_delivery', 'routes', 'vehicle', and 'zone'. The main editor window contains the following SQL query:

```
SELECT zone_id, COUNT(*) AS total_returns FROM deliveries
WHERE is_returned = 1
GROUP BY zone_id ORDER BY total_returns DESC;
```

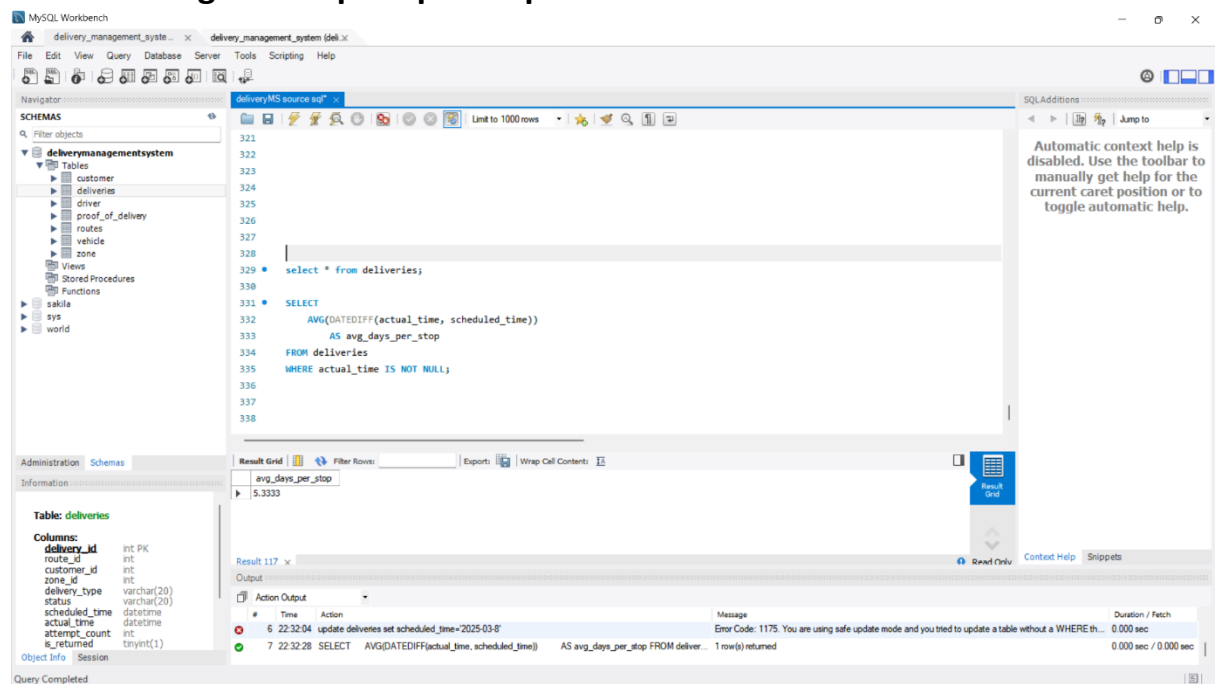
The 'Result Grid' shows the following data:

zone_id	total_returns
2	1

The 'Action Output' pane shows the execution details:

#	Time	Action	Message	Duration / Fetch
2	22:20:25	SELECT zone_id,COUNT(*) AS total_returns from deliveries WHERE is_returned = 1 group by zone_id...	1 row(s) returned	0.000 sec / 0.000 sec
3	22:21:04	SELECT zone_id,COUNT(*) AS total_returns from deliveries WHERE is_returned = 1 group by zone_id...	1 row(s) returned	0.000 sec / 0.000 sec

4. Average time spent per stop.



The screenshot shows the MySQL Workbench interface. The left sidebar displays the 'Schemas' tree with 'deliverymanagementsystem' expanded, showing tables like 'customer', 'deliveries', 'driver', 'proof_of_delivery', 'routes', 'vehicle', and 'zone'. The main editor window contains the following SQL query:

```
SELECT
AVG(DATEDIFF(actual_time, scheduled_time))
AS avg_days_per_stop
FROM deliveries
WHERE actual_time IS NOT NULL;
```

The 'Result Grid' shows the following data:

avg_days_per_stop
5.3333

The 'Table: deliveries' pane shows the following columns:

Column	PK
delivery_id	int PK
customer_id	int
zone_id	int
delivery_type	varchar(20)
status	varchar(20)
scheduled_time	datetime
actual_time	datetime
attempt_count	int
is_returned	tinyint(1)

The 'Action Output' pane shows the execution details:

#	Time	Action	Message	Duration / Fetch
6	22:32:04	update deliveries set scheduled_time='2025-03-8'	Error Code: 1175 You are using safe update mode and you tried to update a table without a WHERE th...	0.000 sec
7	22:32:28	SELECT AVG(DATEDIFF(actual_time, scheduled_time)) AS avg_days_per_stop FROM deliveries	1 row(s) returned	0.000 sec / 0.000 sec

5. Count of deliveries per vehicle type.

The screenshot shows the MySQL Workbench interface. The left sidebar displays the 'deliverymanagementsystem' database schema with tables: customer, deliveries, driver, proof_of_delivery, routes, vehicle, and zone. The main editor contains the following SQL query:

```
select * FROM deliveries ;
select * FROM vehicle;
select * FROM routes;

Select v.vehicle_type, COUNT(d.delivery_id) AS totaldelivery
FROM deliveries d
Join routes r ON d.route_id = r.route_id
JOIN vehicle v ON r.vehicle_id = v.vehicle_id
GROUP BY v.vehicle_type;
```

The 'Result Grid' shows the output of the query:

vehicle_type	totaldelivery
Bike	6
Van	2
Truck	2

The 'Output' pane shows the execution log with the following entries:

#	Time	Action	Message	Duration / Fetch
5	22:37:56	select * FROM routes LIMIT 0, 1000	5 row(s) returned	0.000 sec / 0.000 sec
6	22:41:21	Select v.vehicle_type, COUNT(d.delivery_id) AS totaldelivery FROM deliveries d Join routes r ON d.route_id = r.route_id JOIN vehicle v ON r.vehicle_id = v.vehicle_id GROUP BY v.vehicle_type	3 row(s) returned	0.000 sec / 0.000 sec

Advanced Queries

1. Generate a daily route manifest (Route ID, Driver, Total Stops).

The screenshot shows the MySQL Workbench interface. The left sidebar displays the 'deliverymanagementsystem' database schema with tables: customer, deliveries, driver, proof_of_delivery, routes, vehicle, and zone. The main editor contains the following SQL query:

```
select r.route_id, r.driver_id, count(d.delivery_id) total_stops
from routes r
join deliveries d on r.route_id = d.route_id
where r.route_date = curdate()
group by r.route_id, r.driver_id;
```

The 'Result Grid' shows the output of the query:

route_id	driver_id	total_stops
301	101	1
302	102	1
303	103	2
304	104	1
305	101	5

The 'Output' pane shows the execution log with the following entries:

#	Time	Action	Message	Duration / Fetch
1	23:28:59	select r.route_id, r.driver_id, count(d.delivery_id) total_stops from routes r join deliveries d on r.route_id = d.route_id where r.route_date = curdate() group by r.route_id, r.driver_id	Error Code: 1064. You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near 'count(d.delivery_id) total_stops from routes r join deliveries d on r.route_id = d.route_id where r.route_date = curdate() group by r.route_id, r.driver_id'	0.015 sec
2	23:29:14	select r.route_id, r.driver_id, count(d.delivery_id) total_stops from routes r join deliveries d on r.route_id = d.route_id where r.route_date = curdate() group by r.route_id, r.driver_id	Error Code: 1064. You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near 'count(d.delivery_id) total_stops from routes r join deliveries d on r.route_id = d.route_id where r.route_date = curdate() group by r.route_id, r.driver_id'	0.000 sec
3	23:29:20	select r.route_id, r.driver_id, count(d.delivery_id) total_stops from routes r join deliveries d on r.route_id = d.route_id where r.route_date = curdate() group by r.route_id, r.driver_id	5 row(s) returned	0.000 sec / 0.000 sec

2. List drivers who have worked more than 50 hours this week.

The screenshot shows the MySQL Workbench interface with the 'delivery_management_system' database selected. The 'Schemas' pane on the left shows the database structure. The 'SQL Editor' pane contains the following query:

```
select * from routes;

select driver_id from routes
group by driver_id
having sum(total_distance) > 50;
```

The 'Result Grid' pane shows the results of the first query, listing driver IDs: 101, 102, 103, and 104. The 'Action Output' pane shows the execution details of the second query, indicating that 4 rows were returned.

Table: deliveries

Column	Definition
delivery_id	int PK
route_id	int
customer_id	int
zone_id	int
delivery_type	varchar(20)
status	varchar(20)
scheduled_time	datetime
actual_time	datetime
attempt_count	int
is_returned	tinyint(1)

3. Identify deliveries where the actual time exceeded the estimated time by 30 mins.

The screenshot shows the MySQL Workbench interface with the 'delivery_management_system' database selected. The 'Schemas' pane on the left shows the database structure. The 'SQL Editor' pane contains the following query:

```
select * from deliveries;

select delivery_id, status, scheduled_time, actual_time
from deliveries
where actual_time is not null
and timestampdiff(minute, scheduled_time, actual_time) > 30;
```

The 'Result Grid' pane shows the results of the first query, listing delivery IDs: 1, 3, 5, 7, and 21. The 'Action Output' pane shows the execution details of the second query, indicating that 6 rows were returned.

Column: delivery_id

Definition: delivery_id int PK

4. Find addresses that have been marked "Incorrect Address" multiple times.

The screenshot shows the MySQL Workbench interface. The left sidebar displays the 'Schemas' tree with 'deliverymanagementsystem' expanded, showing tables like 'customer', 'deliveries', 'routes', and 'vehicle'. The central editor contains the following SQL query:

```
select * from deliveries;
select * from routes;
select * from customer;
select c.address, count(*) as attempts from deliveries d
join customer c on d.customer_id = c.customer_id
where lower(d.status) = 'failed'
group by c.address;
```

The 'Result Grid' shows the output of the query:

address	attempts
CS nagar	1
Yercaud	1

The 'Information' tab at the bottom shows the 'Table: customer' structure:

Columns:	
customer_id	int PK
name	varchar(100)
address	varchar(255)
city	varchar(50)
state	varchar(50)
zip_code	varchar(10)
phone	varchar(20)
email	varchar(100)
zone_id	int

The 'Action Output' tab shows the execution progress of the query.

5. Create a view showing: Delivery ID, Customer, Address, Status, and Driver.

The screenshot shows the MySQL Workbench interface. The left sidebar displays the 'Schemas' tree with 'deliverymanagementsystem' expanded, showing tables like 'customer', 'deliveries', 'routes', and 'vehicle'. The central editor contains the following SQL query:

```
select d.delivery_id,
       c.name,
       c.address,
       d.status,
       d.driver_id
from deliveries d
join customer c on d.customer_id = c.customer_id;
```

The 'Result Grid' shows the output of the query:

delivery_id	name	address	status	driver_id
1	Ramesh raj	md colony	Delivered	101
2	Suresh kumar	CS nagar	Failed	102
3	Anitha	Eachanari	Delivered	103
4	Kumar	Yercaud	failed	103
5	Divya	Anna Nagar	Delivered	101

The 'Information' tab at the bottom shows the 'Table: customer' structure:

Columns:	
customer_id	int PK
name	varchar(100)
address	varchar(255)
city	varchar(50)
state	varchar(50)
zip_code	varchar(10)
phone	varchar(20)
email	varchar(100)
zone_id	int

The 'Action Output' tab shows the execution progress of the query.

CREATING INDEX:

The screenshot shows the MySQL Workbench interface with the 'deliveryMS source sq' editor open. The left sidebar displays the 'Schemas' tree with 'deliverymanagementsystem' expanded, showing tables like 'customer', 'deliveries', 'driver', 'proof_of_delivery', 'routes', and 'vehicle'. The 'deliveries' table is selected, and its columns are listed: customer_id, route_id, zone_id, delivery_type, status, scheduled_time, actual_time, attempt_count, is_returned, driver_id, rating, and vehicle_id. The main editor contains the following SQL code:

```
CREATE INDEX idx_routes_route_id
ON routes(route_id);

Create index idx_routes_driver_id
ON routes(driver_id);

CREATE INDEX idx_deliveries_status
ON deliveries(status);
```

The 'Output' pane at the bottom shows the execution results of these commands:

#	Time	Action	Message	Duration / Fetch
1	23:31:34	CREATE INDEX idx_routes_route_id ON routes(route_id)	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.110 sec
2	23:31:56	CREATE INDEX idx_routes_driver_id ON routes(driver_id)	Error Code: 1054. You have an error in your SQL syntax; check the manual that corresponds to your MySQL...	0.000 sec
3	23:32:22	Create index idx_routes_driver_id ON routes(driver_id)	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.063 sec
4	23:32:30	CREATE INDEX idx_deliveries_status ON deliveries(status)	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.063 sec

WITH INDEXING:

The screenshot shows the MySQL Workbench interface with the 'deliveryMS source sq' editor open. The left sidebar displays the 'Schemas' tree with 'deliverymanagementsystem' expanded, showing tables like 'customer', 'deliveries', 'driver', 'proof_of_delivery', 'routes', and 'vehicle'. The 'deliveries' table is selected, and its columns are listed: customer_id, route_id, zone_id, delivery_type, status, scheduled_time, actual_time, attempt_count, is_returned, driver_id, rating, and vehicle_id. The main editor contains the following SQL code:

```
DROP INDEX idx_deliveries_status ON deliveries;

EXPLAIN
SELECT *FROM deliveries
WHERE status = 'Delivered';
```

The 'Output' pane at the bottom shows the execution results of these commands:

#	Time	Action	Message	Duration / Fetch
1	23:37:26	DROP INDEX idx_deliveries_status ON deliveries	Error Code: 1051. Can't DROP 'idx_deliveries_status'; check that column/key exists	0.016 sec
2	23:37:38	EXPLAIN SELECT *FROM deliveries WHERE status = 'Delivered'	1 row(s) returned	0.000 sec / 0.000 sec

The 'Result Grid' pane shows the results of the EXPLAIN query:

id	select_type	table	partitions	type	possible_keys	key	key_len	ref	rows	filtered	Extra
1	SIMPLE	deliveries		ALL					10	10.00	Using where

WITHOUT INDEXING:

The screenshot shows the MySQL Workbench interface. The left sidebar displays the 'Schemas' tree with the 'deliverymanagementsystem' database selected. The 'deliveries' table is highlighted, showing its columns: delivery_id, route_id, customer_id, zone_id, delivery_type, status, scheduled_time, actual_time, attempt_count, is_returned, driver_id, and rating. The main editor window contains the following SQL code:

```
449 CREATE INDEX idx_deliveries_status
450 ON deliveries(status);
451
452 EXPLAIN
453 SELECT *FROM deliveries
454 WHERE status = 'Delivered';
455
456
457
458
459
460
461
462
463
```

The 'Result Grid' shows the execution plan for the query:

id	select_type	table	partitions	type	possible_keys	key	key_len	ref	rows	filtered	Extra
1	SIMPLE	deliveries		ref	idx_deliveries_status	idx_deliveries_status	83	const	7	100.00	

The 'Output' pane shows the execution results:

#	Time	Action	Message	Duration / Fetch
1	23.38.49	CREATE INDEX idx_deliveries_status ON deliveries(status)	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.047 sec
2	23.39.03	EXPLAIN SELECT *FROM deliveries WHERE status = 'Delivered'	1 row(s) returned	0.000 sec / 0.000 sec

RBAC with GRANT / REVOKE:

The screenshot shows the MySQL Workbench interface. The left sidebar displays the 'Schemas' tree with the 'deliverymanagementsystem' database selected. The main editor window contains the following SQL code:

```
461 use deliverymanagementsystem;
462 CREATE USER 'admin_user'@'localhost' IDENTIFIED BY 'admin123';
463 CREATE USER 'dispatcher_user'@'localhost' IDENTIFIED BY 'dispatch123';
464 CREATE USER 'driver_user'@'localhost' IDENTIFIED BY 'driver123';
465 GRANT ALL PRIVILEGES ON deliverymanagementsystem.*
466 TO 'admin_user'@'localhost';
467 GRANT select on deliverymanagementsystem.deliveries
468 TO 'dispatcher_user'@'localhost';
469 GRANT update(route_id, status) ON deliverymanagementsystem.deliveries
470 TO 'dispatcher_user'@'localhost';
471 GRANT SELECT ON deliverymanagementsystem.routes
472 TO 'driver_user'@'localhost';
473 GRANT SELECT ON deliverymanagementsystem.deliveries
474 TO 'driver_user'@'localhost';
475 FLUSH PRIVILEGES;
476 UPDATE deliveries
477 SET status = 'Out for Delivery'
478 WHERE delivery_id = 701;
479
```

The 'Output' pane shows the execution results:

#	Time	Action	Message	Duration / Fetch
1	00:02:46	CREATE USER 'admin_user'@'localhost' IDENTIFIED BY 'admin123'	Error Code: 1396. Operation CREATE USER failed for 'admin_user'@'localhost'	0.015 sec
2	00:02:51	GRANT ALL PRIVILEGES ON deliverymanagementsystem.* TO 'admin_user'@'localhost'	0 row(s) affected	0.000 sec
3	00:02:54	GRANT select on deliverymanagementsystem.deliveries TO 'dispatcher_user'@'localhost'	0 row(s) affected	0.016 sec
4	00:02:58	grant update(route_id, status) ON deliverymanagementsystem.deliveries TO 'dispatcher_user'@'localhost'	0 row(s) affected	0.016 sec
5	00:03:13	GRANT SELECT ON deliverymanagementsystem.routes TO 'driver_user'@'localhost'	Error Code: 1146. Table 'deliverymanagementsystem.routes' doesn't exist	0.000 sec
6	00:03:18	GRANT SELECT ON deliverymanagementsystem.routes TO 'driver_user'@'localhost'	0 row(s) affected	0.015 sec
7	00:03:22	GRANT SELECT ON deliverymanagementsystem.deliveries TO 'driver_user'@'localhost'	0 row(s) affected	0.015 sec
8	00:03:27	FLUSH PRIVILEGES	0 row(s) affected	0.016 sec
9	00:04:09	UPDATE deliveries SET status = 'Out for Delivery' WHERE delivery_id = 701	0 row(s) affected Rows matched: 0 Changed: 0 Warnings: 0	0.000 sec

vulnerable SQL Injection:

The screenshot shows the MySQL Workbench interface with a query window titled 'deliveryMS source.sql'. The query being executed is:

```
SELECT * FROM driver
WHERE license_number = '' OR '1'='1'
```

The result grid displays the following data:

driver_id	first_name	license_number	phone	shift
101	Ramesh	DL-12345	1234567890	Morning
102	Suresh	DL-54321	5432167890	Evening
103	Rajesh	DL-67890	0987654321	Afternoon
104	Sanjay	DL-11111	0987554321	Evening

The 'Output' tab shows the execution details:

#	Time	Action	Message	Duration / Fetch
1	00:06:10	SELECT * FROM driver WHERE license_number = '' OR '1'='1' LIMIT 0, 1000	4 row(s) returned	0.000 sec / 0.000 sec

The 'Table: deliveries' section on the left shows the following columns:

- delivery_id: int PK
- route_id: int
- customer_id: int
- zone_id: int
- delivery_type: varchar(20)
- status: varchar(20)
- scheduled_time: datetime
- actual_time: datetime
- attempt_count: int
- is_returned: tinyint(1)

The 'Administration' tab shows the 'Schemas' section with the following tables:

- customer
- deliveries
- driver
- proof_of_delivery
- routes
- vehicle
- zone
- Views
- Stored Procedures
- Functions
- sys
- world

The 'SQL Additions' panel on the right contains the text: "Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help."